# SDG BLOCKCHAIN ACCELERATOR

## Cohort 2 - Performance Evaluation Report

# 1 Context and Evaluation Approach

The SDG Blockchain Accelerator Cohort 2 was designed to support the development of blockchain-based solutions addressing real-world sustainable development challenges across diverse country contexts. Building on prior program experience, Cohort 2 placed a strong emphasis on execution readiness, institutional alignment, and pilot feasibility.

To ensure a structured and evidence-based evaluation, a standardized assessment framework was implemented, jointly developed by EMURGO Labs and UNDP. This framework evaluates each Solution Maker (SM) and UNDP Challenge Owner (CO) pairing through two complementary perspectives:

- EMURGO Labs Evaluation – focusing on implementation feasibility, technical robustness, and execution maturity
- UNDP Impact & Adoption Evaluation – focusing on SDG alignment, stakeholder engagement, and sustainability potential

All projects were assessed using a quantitative scoring model (1–5 scale) supported by qualitative reviewer comments capturing contextual insights, operational challenges, and deployment readiness.

This dual-layer evaluation ensures that solutions are not only technically viable but also institutionally grounded and capable of real-world adoption.


**SDG BLOCKCHAIN ACCELERATOR** 1
**1 Context and Evaluation Approach** 2
**2 Purpose of the Progress Tracking Document** 3
**3 Document Structure** 3
**4 Cross-Cutting Findings from Cohort 1** 5
**4.1 Hybrid On-Chain / Off-Chain Architectures as a Practical Default** 5
**4.2 Maturing Use of Aiken and UTxO Discipline** 5
**4.3 Regulatory Engagement Began Early and Often** 6
**4.4 Performance, Batching, and Cost Efficiency Are Critical for Scale** 6
**4.5 Institutional Capacity Is Often the Hardest Problem** 7
**5 How This Feeds into Performance Scoring** 7
**6 Cohort 1 Performance Scoring Report** 7
**7 EMURGO Labs – Product Evaluation Scoring** 8
**7.1 Purpose and Methodology** 8

7.2 Evaluation Criteria and Scoring Legend 8
7.3 Detailed Scoring Results 9
7.4 Final Product Scoring Summary 15
7.5 Analytical Insights 16
8. EMURGO Labs – Technical Assessment Scoring 17
8.1 Purpose and Methodology 17
8.2 Evaluation Criteria and Scoring Legend 17
8.3 Detailed Scoring Results 18
8.4 Final Technical Scoring Summary 26
8.5 Analytical Insights 27

# 2 Evaluation Framework

The Cohort 2 evaluation framework was designed to capture both technical feasibility and development impact, recognizing that successful deployment in public-sector contexts requires alignment across multiple dimensions.

Each project was evaluated across five core criteria:

1. **Challenge Alignment, Impact Potential & SDG Contribution** - The extent to which the solution addresses a clearly defined problem and demonstrates measurable alignment with Sustainable Development Goals.
2. **Key Stakeholder Engagement & Partnerships -** The degree of engagement with relevant stakeholders, including UNDP Country Offices, government institutions, NGOs, and private-sector actors.
3. **Implementation / Technical Feasibility** - The practicality of deploying the solution within the local context, considering infrastructure, regulatory requirements, and operational constraints.
4. **Funding Potential & Sustainability** - The existence of a realistic pathway toward long-term sustainability, including funding sources, business models, or institutional support.
5. **Team Commitment & Execution** - The team's ability to execute, including responsiveness, milestone delivery, coordination with partners, and operational discipline.

### Scoring Legend

| Score | Meaning |
|---|---|
| 1 | Very Poor / Not Demonstrated |
| 2 | Weak / Needs Significant Support |
| 3 | Moderate / Shows Potential |
| 4 | Strong / Ready to Engage |
| 5 | Excellent / Fully Aligned and High Potential |

# 3. EMURGO Labs – Evaluation Results

## 3.1 Detailed Scoring Matrix

*Table 1. EMURGO Labs Evaluation – Cohort 2*

| SM-CO Match | Alignment | Stakeholders | Feasibility | Sustainability | Execution | Final Score |
|---|---|---|---|---|---|---|
| ClimaFi <> UNDP Office of Procurement | 3 | 3 | 4 | 4 | 2 | 16.00 |

| | | | | | | |
|---|---|---|---|---|---|---|
| CubID Protocol <> UNDP Nigeria | 4 | 5 | 4 | 4 | 5 | 22.00 |
| Plastiks <> UNDP El Salvador | 4 | 5 | 4 | 4 | 5 | 22.00 |
| ClimaFi <> Better Than Cash Alliance | 4 | 4 | 3 | 4 | 2 | 17.00 |
| Fuel Switch <> UNDP Bosnia and Herzegovina | 5 | 4 | 4 | 4 | 5 | 22.00 |
| TradeChainX <> UNDP Ethiopia | 3 | 5 | 4 | 3 | 3 | 18.00 |
| Green Giraffe Zambia <> UNDP Angola | 4 | 4 | 4 | 3 | 4 | 19.00 |
| Plastiks <> UNDP India | 4 | 5 | 5 | 4 | 3 | 21.00 |
| XCapit <> UNDP Malawi | 4 | 3 | 3 | 3 | 5 | 18.00 |

| | | | | | | |
|---|---|---|---|---|---|---|
| SALA <> UNDP Bosnia and Herzegovina | 4 | 3 | 3 | 4 | 3 | 17.00 |

## 3.2 Final Scoring Summary

*Table 2. Final EMURGO Evaluation Scores*

| Rank | SM-CO Match | Final Score |
|---|---|---|
| 1 | CubID Protocol <> UNDP Nigeria | 22.00 |
| 1 | Plastiks <> UNDP El Salvador | 22.00 |
| 1 | Fuel Switch <> UNDP Bosnia and Herzegovina | 22.00 |
| 4 | Plastiks <> UNDP India | 21.00 |
| 5 | Green Giraffe Zambia <> UNDP Angola | 19.00 |
| 6 | TradeChainX <> UNDP Ethiopia | 18.00 |
| 6 | XCapit <> UNDP Malawi | 18.00 |
| 8 | ClimaFi <> Better Than Cash Alliance | 17.00 |

| | | |
|---|---|---|
| **8** | **SALA <> UNDP Bosnia and Herzegovina** | **17.00** |
| **10** | **ClimaFi <> UNDP Office of Procurement** | **16.00** |

## 3.3 Analytical Insights

### Top Performers

The highest-performing projects, CubID Protocol, Plastiks (El Salvador), and Fuel Switch, demonstrate strong alignment across all evaluation dimensions.

These teams are characterized by:

- Direct engagement with real-world stakeholders and pilot environments
- Strong execution discipline and timely delivery
- Clear alignment with national and sector-specific challenges
- Practical, deployable solutions rather than purely conceptual designs

In particular, CubID Protocol stands out for its extensive field validation and direct engagement with local infrastructure providers, significantly strengthening implementation credibility.

### Strong Performers

Projects such as Plastiks (India) and Green Giraffe Zambia demonstrate solid technical foundations and meaningful alignment with development objectives.

However, further progress is required in:

- Strengthening operational partnerships
- Clarifying scaling and funding pathways
- Advancing toward fully defined pilot implementations

### Mid-Tier Projects

Teams such as TradeChainX and XCapit present promising solutions but remain at an earlier stage of deployment readiness.

Key challenges include:

- Limited on-the-ground validation
- Incomplete stakeholder formalization
- Need for clearer pilot structure and execution planning

### Projects Requiring Further Development

Projects such as ClimaFi and SALA highlight a gap between strong conceptual design and execution readiness.

Observed challenges include:

- Coordination and communication issues with institutional partners
- Misalignment between pilot scope and resource allocation
- Incomplete transition from prototype to deployable solution

These findings reinforce that execution capability is the primary success factor in Cohort 2.

## 3.4 Qualitative Assessment and Reviewer Commentary

In addition to the quantitative scoring framework, each project was evaluated through detailed qualitative feedback provided by EMURGO Labs reviewers. These comments offer deeper insight into execution quality, stakeholder dynamics, architectural decisions, and real-world readiness, complementing the numerical evaluation.

The qualitative assessment highlights not only strengths and achievements, but also critical challenges and recommendations for improvement, providing a more nuanced understanding of each project's maturity and deployment potential.

### Key Observations from Reviewer Feedback

Across the cohort, several recurring themes emerged:

### Strong Conceptual Foundations vs. Execution Gaps

Several projects demonstrated high conceptual clarity and strong alignment with real-world problems, particularly in areas such as climate finance and digital infrastructure.

However, in some cases, execution did not fully match the strength of the initial concept. As noted in the evaluation:

> "The opportunity is strong, but success will depend on execution rigor as much as technical vision."

Challenges included:

- Coordination difficulties with institutional stakeholders
- Misalignment between pilot scope and resource allocation
- Delays in translating design into operational pilots

### Importance of On-the-Ground Validation

Top-performing teams distinguished themselves through direct field engagement and real-world validation.

For example:

> "The team worked closely with the UNDP Country Office to identify real pilot sites... significantly strengthening the credibility of the solution."

This type of engagement enabled:

- Testing assumptions in real environments
- Gathering user feedback early
- Increasing stakeholder trust and adoption readiness

### Execution Discipline as a Differentiator

Reviewer feedback consistently emphasized that execution quality was the primary differentiator across teams.

Projects that:

- Met deadlines
- Delivered working prototypes
- Maintained strong communication with partners

were significantly more likely to achieve higher scores and demonstrate pilot readiness.

Conversely, weaker-performing teams often faced:

- Communication and alignment challenges
- Inconsistent delivery
- Lack of clear execution prioritization

### Shift Toward Deployment and Funding Readiness

Several teams demonstrated a mature understanding that success depends not only on technical delivery but also on funding and deployment pathways.

As highlighted in the feedback:

> "The priority was not adding more features, but unlocking adoption and scaling pathways."

This reflects a broader shift toward:

- Partnership development
- Securing co-financing
- Preparing for real-world rollout

### Institutional Coordination as a Key Bottleneck

A recurring challenge across multiple projects was the complexity of working within institutional environments.

Reviewer observations indicate that:

- Misalignment within Country Offices can delay execution
- Lack of clearly defined focal points impacts progress
- Coordination overhead can slow otherwise strong solutions

These findings reinforce that institutional readiness is as critical as technical readiness.

### Transition from Proof-of-Concept to Production

Several projects are currently in transition from PoC-stage solutions to deployable systems, particularly those involving advanced architectures (e.g., identity systems, financial infrastructure).

Reviewer feedback highlights that this transition requires:

- Stronger system integration (APIs, storage, performance testing)
- Completion of architectural migrations
- Validation under real-world conditions

### Summary

The qualitative assessment reinforces the core finding of Cohort 2:

Successful projects are those that combine strong conceptual design with disciplined execution and real-world validation.

While many teams demonstrate high potential, the ability to:

- Coordinate effectively with stakeholders
- Deliver consistently
- Translate concepts into operational pilots

remains the defining factor in achieving deployment readiness.

### 3.5 Project-Level Qualitative Evaluation

In addition to the aggregated insights presented above, detailed reviewer comments were analyzed at the level of individual SM–CO pairings. These project-level assessments provide a more granular understanding of each solution's strengths, limitations, and readiness for pilot deployment.

***Table 3. Reviewer Commentary by Project – Cohort 2***

| SM–CO Match | Reviewers' Comment |
|---|---|
| **ClimaFi <> UNDP Office of Procurement** | ClimaFi demonstrates a strong conceptual foundation, addressing a critical gap in climate finance infrastructure with a well-articulated and relevant use case. The team shows a solid understanding of both technical and institutional dimensions, particularly through its pivot toward a digital audit and shadow payment model.<br><br>However, execution has not fully matched the strength of the concept. Key challenges were observed in coordination and alignment with institutional stakeholders, which is particularly critical given the nature of the solution. Additionally, the proposed pilot scope and budget appeared misaligned, |

| | |
|---|---|
| | raising concerns around prioritization and cost-efficiency.<br><br>Moving forward, improved stakeholder coordination and more disciplined scoping will be essential to translate conceptual strength into a viable pilot. |
| **CubID Protocol <> UNDP Nigeria** | CubID stands out as one of the strongest and most well-executed projects in the cohort. The team successfully combines technical innovation with practical implementation, notably through the use of Hydra for real-time settlement.<br><br>A key differentiator is the team's strong field engagement. Direct collaboration with the UNDP Nigeria Country Office, solar mini-grid operators, and on-the-ground validation significantly strengthens the solution's credibility and deployability.<br><br>The project demonstrates a high level of execution maturity, positioning it well for scaling, provided that the same level of operational discipline is maintained. |
| **Plastiks <> UNDP El Salvador** | Plastiks represents a clear example of execution-focused maturity. Rather than expanding technically, the team prioritized real-world deployment, stakeholder alignment, and funding acquisition.<br><br>The successful securing of external funding and transition into an active pilot phase |

| | |
|---|---|
| | reflects strong alignment between solution design and implementation strategy. The team demonstrates a clear understanding that scaling depends on adoption pathways rather than additional feature development.<br><br>This positions Plastiks as one of the most advanced projects in terms of real-world readiness. |
| **ClimaFi <> Better Than Cash Alliance** | This implementation of ClimaFi demonstrates strong institutional alignment and relevance within financial inclusion and climate finance ecosystems. Stakeholder positioning is well developed, providing a solid foundation for future adoption.<br><br>However, similar challenges to the other ClimaFi track are observed, particularly regarding execution discipline and pilot scoping. The proposed budget remains disproportionate to the scope of validation, and coordination challenges have impacted progress.<br><br>Addressing these issues will be critical for moving from conceptual alignment to operational deployment. |
| **Fuel Switch <> UNDP Bosnia and Herzegovina** | Fuel Switch delivered a strong and consistent performance throughout the accelerator. The team met all key milestones, developed a pilot-ready platform, and successfully |

| | |
|---|---|
| | engaged both government and private-sector stakeholders.<br><br>The solution addresses a clearly defined and time-sensitive problem (CBAM compliance), enhancing its relevance and adoption potential. Execution quality and stakeholder alignment are key strengths.<br><br>The primary outstanding challenge relates to securing funding for the next phase, which depends on external support. Overall, the project demonstrates high readiness for continued development. |
| **TradeChainX <> UNDP Ethiopia** | TradeChainX has made solid progress in shaping its solution and engaging with relevant stakeholders, laying a foundation for future pilot implementation.<br><br>However, the project remains at an earlier stage compared to higher-performing teams. Further work is needed to translate the concept into a clearly defined and testable pilot, including stronger stakeholder commitments, defined data flows, and measurable success indicators.<br><br>Continued focus on operationalization and field validation will be essential. |
| **Green Giraffe Zambia <> UNDP Angola** | Green Giraffe successfully delivered a technically sound proof-of-concept, demonstrating strong capabilities in traceability and token lifecycle implementation on Cardano. |

| | |
|---|---|
| | The solution is well aligned with SDG priorities, particularly in agricultural supply chain transparency. However, the project has not yet established sufficient real-world partnerships or a clearly defined business model.<br><br>Further efforts are required to transition from a technical PoC to a deployable pilot involving actual stakeholders such as producers and off-takers. |
| **Plastiks <> UNDP India** | This implementation demonstrates strong interest and engagement from the UNDP India Country Office, reflecting high relevance of the solution within the local context.<br><br>However, progress has been impacted by coordination challenges within the institutional environment. While such challenges are common, they highlight the importance of clearly defined focal points and streamlined communication channels.<br><br>The identification of a more aligned stakeholder within the Country Office represents a positive step toward improving execution momentum. |
| **XCapit <> UNDP Malawi** | XCapit presents a highly mature and technically advanced solution, particularly in addressing low-connectivity environments through SMS-based infrastructure.<br><br>The team demonstrates a strong understanding of local challenges and has |

| | |
|---|---|
| | already validated its solution in other contexts. However, the main limitation lies in incomplete institutional anchoring.<br><br>While the solution is technically ready, successful deployment will depend on formalizing partnerships, defining pilot roles, and securing stronger engagement from local stakeholders. |
| **SALA <> UNDP Bosnia and Herzegovina** | SALA has undergone a significant evolution during the accelerator, transitioning from a basic NFT-based approach to a more robust architecture based on decentralized identity (DID) and verifiable credentials.<br><br>This shift reflects a strong learning curve and improved alignment with real-world requirements. Collaboration with academic institutions and alignment with emerging standards are positive indicators.<br><br>However, the solution remains in a proof-of-concept stage. Key components such as system integration, performance validation, and full architectural migration are still pending.<br><br>Further development is required to reach pilot-ready maturity. |

The project-level qualitative evaluation highlights a clear pattern across the cohort:

- Top-performing projects combine technical capability with strong execution and stakeholder alignment
- Mid-tier projects require further operational grounding and validation
- Lower-performing projects demonstrate strong concepts but lack execution maturity

These insights reinforce the broader conclusion that successful deployment depends not only on innovation, but on the ability to operationalize solutions within real institutional environments.

## 4. Descriptive Statistics and Distribution

**The Cohort 2 results indicate a relatively consistent performance distribution across projects.**

***Table 4. Descriptive Statistics – Cohort 2***

| Metric | Value |
|---|---|
| Mean Score | 19.20 |
| Standard Deviation | 2.04 |
| Minimum Score | 16.00 |
| Maximum Score | 22.00 |

The relatively low standard deviation suggests that Cohort 2 achieved greater consistency in execution readiness, reflecting improved selection processes and clearer program expectations.

## 5. Interpretation of Results

Cohort 2 represents a clear evolution toward deployment-oriented innovation, where success is defined by the ability to transition from concept to real-world implementation.

**Execution as the Primary Differentiator**

Projects that performed best were not necessarily the most technically complex, but those that:

- Demonstrated consistent delivery
- Maintained strong stakeholder coordination
- Focused on realistic and achievable pilot outcomes

**Institutional Alignment as a Critical Factor**

Strong performance is closely linked to:

- Active engagement with UNDP Country Offices
- Government involvement and policy alignment
- Clearly defined roles and responsibilities for pilot implementation

**Transition from Prototype to Pilot**

A defining characteristic of Cohort 2 is the shift toward:

- Real pilot environments
- Secured funding pathways
- Operational testing in real-world conditions

This represents a significant step toward establishing blockchain solutions as practical tools for public-sector innovation.

# 6. Lessons Learned

## 6.1 Early Stakeholder Commitment is Essential

Projects with early institutional engagement demonstrated significantly higher readiness for deployment.

## 6.2 Realistic Scoping Drives Execution Success

Solutions with clearly defined and achievable pilot scopes were more successful than those with overly ambitious designs.

## 6.3 Execution Discipline Outweighs Conceptual Strength

Consistent delivery, coordination, and responsiveness proved to be stronger predictors of success than conceptual innovation alone.

### 6.4 Field Validation is a Key Success Factor

Projects that engaged directly with end users and real-world environments demonstrated higher credibility and readiness.

### 6.5 Sustainability Must Be Addressed Early

Clear funding pathways and operational models are critical for transitioning from pilot to scale.

## 7. Concluding Reflections

Cohort 2 demonstrates that blockchain innovation in development contexts is entering a more mature phase, where the focus shifts from experimentation to execution and deployment.

The most successful projects combined:

- Clear problem–solution alignment
- Strong institutional engagement
- Practical implementation strategies
- Measurable pathways to impact

At the same time, the cohort highlights that institutional coordination, execution discipline, and realistic planning remain the primary challenges in achieving scalable deployment.

Looking forward, the SDG Blockchain Accelerator is well-positioned to further strengthen its impact by:

- Standardizing execution frameworks
- Enhancing institutional integration
- Supporting teams in transitioning from pilot to scale

Cohort 2 confirms that the pathway to impact lies not only in innovation, but in the ability to deliver solutions that are usable, adoptable, and sustainable in real-world environments.